

Agile Methodology Implementation

To make sure the LMS project for B&T Skills Development Trainings is delivered efficiently, flexibly, and in a structured way, we will follow an **Agile methodology**. This approach focuses on continuous collaboration, incremental progress, and the ability to adapt to changing requirements as the project develops.

Team Roles in Agile

- **NE Nyembe** – Acting as Business Analyst, System Tester, and Project Manager, NE will gather and prioritise requirements, coordinate sprints, track progress, and ensure quality through testing. NE will also serve as the link between stakeholders and the development team, translating business needs into practical tasks.
- **ET Nthani** – Responsible for Software Development (Programming) and Database Development & Administration, ET will handle the implementation of system functionalities, develop both front-end and back-end solutions, and manage database design, integration, and maintenance. ET works closely with NE to ensure that the technical side aligns with business needs and user expectations.

Agile Practices

- **Sprint Planning:** The team will plan short development cycles, or sprints, delivering small, manageable features each time.
- **Daily Standups:** Short daily meetings will help the team stay coordinated, remove obstacles, and monitor progress.
- **Backlog Management:** All requirements will be kept in a product backlog and prioritised according to business value and urgency.
- **Continuous Testing:** NE will test the system at the end of each sprint to make sure everything is working correctly and meets quality standards.
- **Sprint Review and Retrospective:** Completed work will be presented to stakeholders for feedback, and the team will review each sprint to identify areas for improvement.

Introduction

This document outlines the Business Analysis for the Learning Management System (LMS) developed for B&T Skills Development Trainings. The main purpose of this system is to provide a centralised digital platform that supports the effective delivery, management, and monitoring of learnership and internship programmes within the organisation.

The LMS is a web-based system that allows different users, including **learners, interns, facilitators, mentors, assessors, moderators, and administrators**, to interact within a secure and structured online environment. Each user role will have specific responsibilities and access to role-based dashboards that support their activities within the training and assessment process.

Instead of relying on manual processes and paperwork, the system introduces automated workflows that make daily academic, administrative, and workplace training tasks more efficient and organised. The platform will also support structured communication between learners, interns, mentors, facilitators, assessors, and moderators to ensure smooth coordination throughout the training programmes.

By implementing this digital solution, B&T Skills Development Trainings aims to strengthen operational efficiency, improve communication between stakeholders, enhance learner and intern engagement, and ensure that all academic and workplace learning data is stored and managed accurately. This initiative also supports the organisation's broader move towards digital transformation within the South African skills development sector while improving compliance with learnership and internship programme requirements.

Background

B&T Skills Development Trainings is a training provider that offers structured learnership and internship programmes aimed at equipping students with practical skills and recognised competencies. These programmes combine theoretical learning with workplace-based training to ensure that learners and interns gain both academic knowledge and practical industry experience.

As the organisation continues to grow and enrol more learners and interns across different programmes, managing operations manually has become increasingly challenging and time-consuming. The training process also involves multiple stakeholders such as facilitators, mentors, assessors, moderators, and administrators, all of whom play different roles in supporting learner development and assessment.

At present, several key activities such as learner and intern registration, assessment submissions, tracking academic and workplace performance, and general communication are either handled manually or through separate, unconnected systems. Mentors often monitor workplace learning outside the system, while assessors and moderators manage evaluation and quality assurance processes separately. This

approach often results in delays, unnecessary paperwork, difficulty in accessing information quickly, and a higher risk of losing important data. It also makes reporting more complicated and less transparent.

To improve service delivery and remain competitive within South Africa's skills development sector, B&T Skills Development Trainings recognised the need for a centralised Learning Management System. The LMS will act as the organisation's digital backbone, bringing together course content, workplace learning activities, assessments, learner records, mentor feedback, and reporting into one structured and reliable platform.

By integrating these functions into a single system, the organisation will improve coordination between learners, interns, facilitators, mentors, assessors, moderators, and administrators while ensuring that all training activities are properly monitored and recorded.

Problem Statement

B&T Skills Development Trainings is currently facing operational challenges due to the reliance on manual processes and systems that are not properly integrated. Learner and intern information is not stored in one central location, making it difficult to accurately track academic and workplace learning progress or generate reliable reports when required.

Assessment submissions and workplace learning evidence are not always monitored consistently, which can result in delayed feedback to learners and interns. This affects the overall learning experience and makes it more difficult to identify participants who may require additional academic or practical support.

Facilitators also experience challenges when managing multiple learner groups without a structured digital system. In addition, mentors who supervise interns in workplace environments often track practical learning activities outside the current system. Assessors and moderators responsible for evaluating learner competence and ensuring quality assurance also face difficulties accessing complete records of learner progress and assessment evidence.

Tasks such as tracking attendance, marking assessments, reviewing portfolios of evidence, and monitoring programme performance require a significant amount of manual effort. Communication between learners, interns, facilitators, mentors, assessors, moderators, and administrative staff is not always well coordinated, which can lead to misunderstandings, delays, and gaps in information.

Without an integrated system in place, the organisation risks reduced productivity, increased administrative errors, and limited ability to effectively manage and scale its training programmes. There is therefore a clear and urgent need for a digital solution that centralises operations, improves efficiency, and supports structured and effective management of learnership and internship programmes.

Objectives

The main goal of this project is to design and implement a fully functional Learning Management System (LMS) that supports both the operational and academic needs of B&T Skills Development Trainings. The system is intended to strengthen how the organisation manages its learnership and internship programmes while improving the overall learning and training experience for learners and interns.

The specific objectives of the system are as follows:

- To centralise learner and intern registration and profile management so that all participant information is stored securely in one place.
- To provide digital access to study materials and training content, allowing learners and interns to access resources anytime and from anywhere.
- To enable online formative and summative assessments, making submissions, marking, and competency evaluations more efficient and structured.
- To support workplace-based learning by allowing mentors to monitor intern progress, provide feedback, and review workplace learning activities.
- To provide real-time tracking of learner and intern performance so that facilitators, assessors, and management can monitor progress and identify areas that require support.
- To support structured assessment and quality assurance processes by allowing assessors to evaluate learner submissions and moderators to verify assessment outcomes.
- To improve communication between learners, interns, facilitators, mentors, assessors, moderators, and administrators through notifications, announcements, and messaging features within the system.
- To generate automated and accurate reports to support management decision-making, programme monitoring, and compliance with SETA and accreditation requirements.
- To strengthen data security and ensure that all academic, workplace learning, and personal information is protected.
- To design the system in a way that allows for future growth as learner numbers, internship placements, and programme offerings increase.

The LMS is aimed at improving administrative efficiency while also creating a more organised, transparent, and supportive learning and training environment for all stakeholders involved in the programmes.

Proposed Solution

The proposed solution is a custom-built, web-based Learning Management System designed specifically to meet the needs of B&T Skills Development Trainings. Instead of using a generic platform, this system is developed to align with the organisation's internal processes, training requirements, and reporting standards. Access to the system will be controlled through secure login credentials, with clearly defined user

roles such as **Administrator, Facilitator, Learner, Intern, Mentor, Assessor, and Moderator.**

Through the system, administrators will be able to register learners and interns, allocate them to the relevant courses or programmes, and monitor overall programme progress from a central dashboard. Facilitators will have the tools they need to upload study materials, create and manage assessments, mark submissions, and provide structured feedback to learners.

Learners will be able to access learning materials, complete assessments online, track their academic performance, and receive important notifications and updates. Interns will have a dedicated dashboard focused on workplace learning activities, where they can submit internship reports, track practical tasks, and receive feedback from mentors.

Mentors will be able to monitor the progress of interns in the workplace, review practical learning activities, and provide feedback on workplace performance. Assessors will evaluate learner submissions and practical evidence to determine competency levels, while moderators will review assessment outcomes to ensure that evaluation processes meet quality assurance and accreditation standards.

The LMS will also include role-based dashboards that provide real-time statistics, progress indicators, and programme insights. These dashboards will support effective monitoring of learner and intern performance while assisting management in making informed decisions and meeting compliance and reporting requirements.

By implementing this solution, the organisation will significantly reduce paperwork, ease the administrative burden on staff, improve the accuracy of academic and workplace learning records, and move confidently towards digital transformation within the training environment.

System Personas

To ensure that the LMS meets the needs of all users, the following personas have been identified. These personas reflect the primary users and their requirements for an effective learning and workplace training experience.

Learner Persona

- Learners are typically young adults enrolled in learnership programmes, often combining theoretical study with practical workplace training.

Facilitator Persona

- Facilitators are responsible for delivering course content and evaluating learner performance.

Administrator Persona

- Administrators oversee the overall operation of the LMS, including learner enrolment, course allocation, and system management.

Intern Persona

- Interns are participants gaining practical industry experience through internship programmes.

Mentor Persona

- Mentors are workplace professionals who guide learners or interns during practical training.

Assessor Persona

- Assessors are qualified professionals responsible for evaluating learner competence in theoretical and practical components.

Moderator Persona

- Moderators ensure quality assurance and verification of assessments, guaranteeing fair, valid, and compliant evaluation outcomes.

Stakeholders Involved

Primary Stakeholders (Direct Users)

These are the main users who interact directly with the LMS to perform learning, training, or operational tasks:

- **Learnership Students** – require easy access to study materials, online assessments (formative and summative), performance tracking, notifications, and academic progress reports. They rely on the system for learning, assignment submission, and receiving timely feedback.
- **Interns** – require access to internship programme details, workplace tasks, mentor evaluations, logbook submissions, and progress tracking. They depend on the system to coordinate workplace learning and receive guidance from mentors and assessors.
- **Facilitators / Trainers** – need tools to upload course content, create and manage assessments, grade submissions, track learner and intern progress, and communicate effectively with students and interns.
- **Mentors** – responsible for guiding interns or learners during workplace activities. They require dashboards to monitor practical progress, provide evaluations, approve logbooks, and communicate with assessors and administrators.

- **Assessors** – require access to learner and intern portfolios, assessment grading tools, competency tracking, and communication channels with mentors and moderators to ensure accurate evaluation.
- **Moderators** – responsible for quality assurance of assessments. They need access to assessment records, learner portfolios, moderation reporting tools, and communication with assessors and management to maintain compliance.
- **Training Coordinators** – require oversight of learner and intern enrolment, programme allocation, attendance tracking, and performance monitoring to ensure smooth academic operations.
- **Management (B&T Leadership Team)** – require access to dashboards and analytical reports to monitor programme performance, learner and intern success rates, compliance requirements, and overall institutional efficiency.

Secondary Stakeholders (Support & Administration)

These stakeholders ensure the LMS operates smoothly and meets technical, operational, and business objectives:

- **System Administrator** – manages user accounts, assigns roles and permissions, updates course information, maintains system configurations, and ensures data accuracy and security.
- **NE Nyembe (Business Analyst, Project Manager & System Tester)** – responsible for requirements gathering, stakeholder engagement, project coordination, quality assurance testing, and ensuring the system aligns with business objectives.
- **ET Nthani** – responsible for software development and database administration, including front-end and back-end implementation, database design, integration, and system maintenance.
- **IT Support Team** – provides technical assistance, monitors system uptime, manages backups, ensures cybersecurity protection, and maintains infrastructure reliability.

External Stakeholders

These stakeholders have an indirect role but are essential for regulatory compliance, system support, and service continuity:

- **SETA and Accreditation Bodies** – require accurate reporting, learner and intern progress documentation, and compliance records for skills development and funding purposes.
- **Service Providers** – may provide hosting services, cloud infrastructure, cybersecurity tools, or third-party integrations required for system performance.
- **Network Providers** – ensure stable and reliable internet connectivity for uninterrupted LMS access.
- **Regulatory and Compliance Authorities** – ensure the LMS complies with data protection regulations, privacy standards, and information security requirements.

Functional Requirements

Functional Requirements for User and Admin Role

ID	Feature / Function	Description
01	User Registration (Sign Up)	The system must allow learners, interns, facilitators, mentors, assessors, moderators, and administrators to create an account by providing required personal and institutional details. Account activation must be verified through a secure email verification process.
02	User Login	The system must allow registered users to log in securely using valid credentials such as email/username and password.
03	Forgot Password	Users must be able to reset their password through a secure password recovery link sent to their registered email address.
04	Profile Management	Users must be able to view and update their personal profile information, including contact details and profile settings.
05	Dashboard Access	The system must provide a role-based dashboard displaying relevant information such as courses, workplace tasks, notifications, system activities, and evaluation milestones.

Functional Requirements for Learner Role

ID	Feature / Function	Description
01	Course Enrolment	The system must allow learners to view assigned courses and access enrolled programme details.
02	Access Study Materials	The system must allow learners to download or view uploaded study materials such as PDFs, presentations, and learning resources.
03	Online Assessment Submission	The system must allow learners to complete and submit formative and summative assessments electronically before the due date.
04	View Results & Feedback	The system must display assessment results and facilitator feedback once grading is completed.
05	Notifications & Announcements	The system must notify learners about upcoming assessments, deadlines, announcements, and course updates.
06	Progress Tracking	The system must allow learners to track their academic performance and completion status within each course.

07	Feedback Submission	Learners must be able to submit feedback or queries related to courses or system functionality.
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Functional Requirements for Intern Role

ID	Feature / Function	Description
01	Internship Access	Interns can view internship programme details, workplace learning tasks, and schedules.
02	Workplace Logbook Submission	Interns can submit workplace learning logs, reports, and practical evidence for mentor and assessor review.
03	Mentor Feedback	Interns receive structured feedback and evaluations from mentors and assessors.
04	Progress Tracking	Interns can monitor their progress through practical tasks, logbook completion, and evaluation milestones.
05	Notifications & Communication	Interns are notified of upcoming tasks, deadlines, and receive messages from mentors or assessors.

Functional Requirements for Facilitator Role

ID	Feature / Function	Description
01	Course Content Management	Facilitators must be able to upload, update, or remove study materials and learning resources.
02	Assessment Creation	Facilitators must be able to create formative and summative assessments with defined deadlines and grading criteria.
03	Grade Submissions	Facilitators must be able to review learner submissions, assign marks, and provide feedback.
04	Performance Monitoring	Facilitators must be able to monitor learner progress, attendance, and overall academic performance.
05	Communication Tool	Facilitators must be able to send announcements or messages to learners within their assigned courses.

Functional Requirements for Mentor Role

ID	Feature / Function	Description
01	Mentor Dashboard	Mentors can view assigned interns/learners and monitor workplace tasks.
02	Task Approval & Feedback	Mentors can approve workplace learning logs and provide structured feedback.
03	Progress Monitoring	Mentors can track practical progress and evaluate completion milestones.
04	Communication with Assessors	Mentors can share learner/intern performance updates and workplace evidence with assessors.

Functional Requirements for Assessor Role

ID	Feature / Function	Description
01	Assessment Review	Assessors can review learner and intern submissions, including theoretical and practical evidence.
02	Competency Evaluation	Assessors can grade assessments and workplace tasks against programme outcomes.
03	Feedback Provision	Assessors can provide structured feedback to learners, interns, and mentors.
04	Communication with Moderators	Assessors can submit evaluation results for quality assurance and moderation.

Functional Requirements for Moderator Role

ID	Feature / Function	Description
01	Assessment Verification	Moderators can review assessor grading and verify compliance with assessment standards.
02	Quality Assurance Reporting	Moderators can generate reports and provide feedback for reassessment if required.
03	Compliance Monitoring	Moderators ensure that assessments follow accreditation requirements and standards.
04	Communication with Admin	Moderators coordinate with administrators and management to maintain assessment compliance.

Functional Requirements for Admin Role

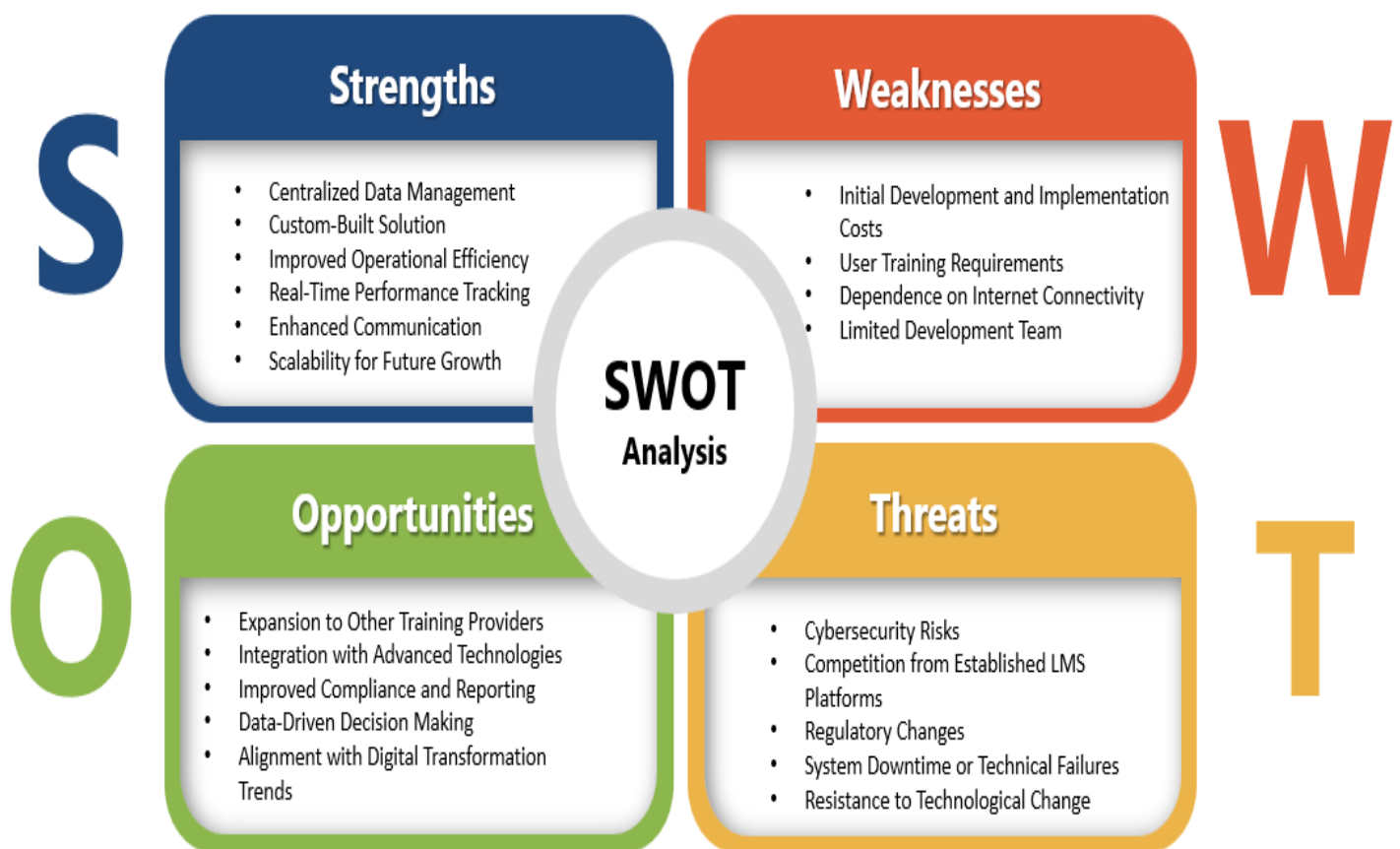
ID	Feature / Function	Description
01	User Management	Admin can create, update, deactivate, or remove user accounts and assign appropriate roles.
02	Course Management	Admin can create, update, assign, or delete courses and allocate facilitators, learners, and interns.
03	Programme Management	Admin can manage learnership and internship programmes, including intake periods and course structures.
04	Reports Dashboard	Admin can generate and download reports related to learner and intern performance, course completion rates, and system usage.
05	System Configuration	Admin can configure system settings, academic calendars, and assessment policies.
06	Database Management	The system must securely store all user data, course materials, assessments, results, and activity logs in a centralized database.

Non-Functional Requirements

ID	Category	Requirement Description
01	Performance	The system must load dashboards and course content within 3 seconds under normal operating conditions and support multiple concurrent users without performance degradation.
02	Usability	The LMS interface must be intuitive, user-friendly, and easy to navigate for learners, facilitators, and administrators with minimal training.
03	Security	All user credentials and sensitive information must be encrypted and securely stored. The system must implement role-based access control to prevent unauthorized access.
04	Reliability	The system must achieve at least 99% availability during operational hours and include automated backup and recovery mechanisms.
05	Scalability	The system must be capable of handling an increasing number of learners, courses, and assessments without affecting performance.

06	Compatibility	The LMS must be accessible through modern web browsers and be responsive across desktop, tablet, and mobile devices.
07	Maintainability	The system architecture must support easy updates, bug fixes, and future enhancements without disrupting active users.
08	Data Integrity	Learner records, assessment results, and course data must remain accurate, consistent, and protected from unauthorized modification.
09	Privacy	The system must comply with data protection regulations, including POPIA, to ensure confidentiality of personal and academic information.
10	Accessibility	The LMS should support accessibility features such as readable fonts, structured layouts, and compatibility with assistive technologies where possible.

SWOT Analysis



SWOT Analysis of the LMS for B&T Skills

Development Trainings

1. Strengths (Internal Positive Factors)

- **Custom-Built Solution** – The LMS is specifically developed for B&T, aligning with operational processes, reporting requirements, and academic structures. It avoids unnecessary features present in generic platforms, improving efficiency and user experience.
- **Centralized Data Management** – All learner, intern, facilitator, mentor, assessor, and moderator data, along with assessments, course materials, and reports, are stored securely in a centralized database. This reduces duplication, fragmentation, and errors while improving reporting accuracy.
- **Improved Operational Efficiency** – Automation of registration, course enrolment, assessment submissions, grading, and reporting reduces manual administrative tasks, freeing staff to focus on academic quality and learner support.
- **Real-Time Performance Tracking** – Facilitators, mentors, and management can monitor learner and intern progress in real time, quickly identifying struggling users and implementing early interventions.
- **Enhanced Communication** – Built-in notifications, announcements, and messaging improve communication between all users (learners, interns, facilitators, mentors, assessors, and administrators), reducing misunderstandings and ensuring important updates are delivered promptly.
- **Scalability for Future Growth** – The LMS can support increasing numbers of users, courses, and programmes as B&T expands, avoiding the need for full redevelopment.

2. Weaknesses (Internal Limitations)

- **Initial Development and Implementation Costs** – Building a custom LMS requires investment in time, technical expertise, and infrastructure. While beneficial long-term, initial costs may strain resources.
- **User Training Requirements** – Learners, interns, facilitators, and mentors may require guidance to effectively use the system. Limited digital literacy or resistance to change can slow adoption.
- **Dependence on Internet Connectivity** – As a web-based platform, the LMS requires stable internet. Learners in low-connectivity areas may face difficulties accessing materials or submitting assessments.
- **Limited Development Team** – With a small team (Business Analyst and Full Stack Developer), updates, enhancements, and maintenance may take longer compared to larger IT departments.

3. Opportunities (External Positive Factors)

- **Expansion to Other Training Providers** – The LMS could be adapted and offered to other institutions, creating potential revenue streams and strategic partnerships.
- **Integration with Advanced Technologies** – Future enhancements like mobile apps, AI-driven performance analytics, automated attendance tracking, or e-certification could improve competitiveness and user experience.
- **Improved Compliance and Reporting** – Automated reporting enhances adherence to SETA and accreditation standards, increasing credibility and funding opportunities.
- **Data-Driven Decision Making** – The LMS generates insights on learner and intern performance, course effectiveness, and completion rates, enabling better curriculum design and strategic planning.
- **Alignment with Digital Transformation Trends** – Adoption of the LMS positions B&T as a modern, forward-thinking institution in the skills development sector.

4. Threats (External Risks)

- **Cybersecurity Risks** – Digital platforms are vulnerable to cyberattacks, data breaches, or unauthorized access. Protecting sensitive learner and staff information is critical.
- **Competition from Established LMS Platforms** – Well-known LMS providers with advanced features and brand recognition could compete with the custom-built system if external comparisons are made.
- **Regulatory Changes** – Updates in data protection laws or education compliance standards may require system modifications. Failure to adapt could result in penalties or non-compliance.
- **System Downtime or Technical Failures** – Server issues, bugs, or maintenance downtime may disrupt academic activities, requiring reliable backup and maintenance strategies.
- **Resistance to Technological Change** – Some staff or learners may prefer traditional methods, slowing LMS adoption and reducing effectiveness.